

2)

Known: thickness of wall A, Wall B thickness of L_B and thermal conductivity, right surface of wall B is exposed to coolant at $T_\infty = 20^\circ\text{C}$, convective heat transfer coefficient of $500\text{ W/m}^2\text{K}$, thermal insulation and wall A temperature, the sensor interface temp,

Find:

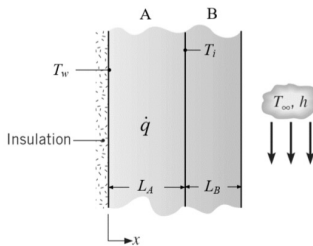
- right surface temp T_s of wall B.
- uniform volumetric heat generation $\dot{q}$ at S.S.
- calculate thermal conductivity k_A of wall A.
- Plot temp profile across wall A & wall B

Assumptions:

one dimensional heat conduction, left surface of wall A is thermally insulated, there is no heat generation within wall B, The left surface of wall B is in contact with wall A, there is no contact thermal resistance at their boundary

Properties: gHS

Schematic:



Analysis:

$$a) \quad \dot{E}_{in} = \dot{E}_{out}$$

$$\begin{aligned} q_x'' &= -k \frac{T_2 - T_1}{L} \\ &= -k_B \frac{T_s - T_i}{L_B} \\ &= k_B \frac{T_i - T_s}{L_B} = h(T_s - T_\infty) \end{aligned}$$

$$(30) \frac{(373 - T_s)}{0.1\text{ m}} = (500)(T_s - 293) =$$

$$T_s = 323\text{ K} \Rightarrow T_s = 323\text{ K} - 273 = \boxed{50^\circ\text{C} = T_s}$$

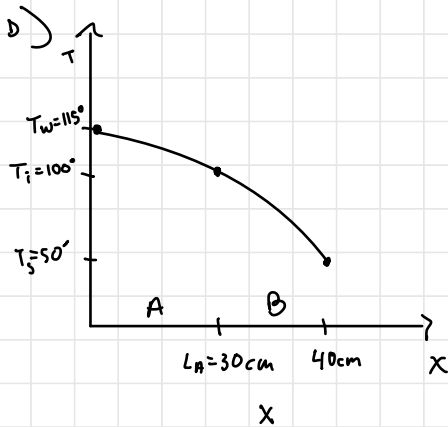
b)

$$\dot{q}'' = \dot{q} L_B$$

$$500(50-20) = \dot{q}(0.30)$$

$$15000 = \dot{q}(0.30)$$

$$\dot{q} = 50,000 \text{ W/m}^3$$



c)

$$\frac{\partial}{\partial x} \left[k \frac{\partial T}{\partial x} \right] + \dot{q} = 0 \quad \left\{ \begin{array}{l} T(L_A, t) = T_i \\ T(0, t) = T_w \\ \frac{\partial T(0)}{\partial x} = 0 \end{array} \right.$$

$$k \frac{\partial^2 T}{\partial x^2} = -\dot{q}$$

$$\frac{\partial^2 T}{\partial x^2} = -\frac{\dot{q}}{k}$$

$$T(x) = C_1 + C_2 x - \frac{\dot{q}}{2k} x^2 = T_w - \frac{\dot{q}}{2k} x^2$$

$$T'(x) = C_2 - \frac{\dot{q}}{2k} x^2$$

$$0 = C_2$$

$$T(0) = T_w$$

$$T_i = T_w - \frac{\dot{q}}{2k} L_A^2$$

$$k = \frac{\dot{q} L_A^2}{2(T_w - T_i)}$$

$$k = 150 \text{ W/mK}$$